

SIMATS ENGINEERING

CO3- ASSESSMENT TOOL 2

SIMULATION TASK

COURSE NAME: DIGITAL SIGNAL PROCESSING

COURSE CODE: ECA09

COURSE OUTCOME COVERED

CO 3: Evaluate finite word-length effects and multirate signal processing techniques, including decimation, interpolation, and polyphase decomposition. **Bloom Level mapped-BL3,BL4,BL5**

Topics Covered:

Quantization- Truncation and Rounding errors - Quantization noise - Limit cycle oscillations – Signal scaling.Parametric and Non parametric Spectral estimation. Decimation-Interpolation-Sampling rate conversion-Implementation of sampling rate conversion-Polyphase Decomposition -5G Decimation/Interpolation: For mmWave beamforming.

Assessment Tool number : CO3-AT2

Assessment Tool Weightage : 35%

Assessment Tool Name : Simulation Task

Duration of this assessment : 60 Minutes

Marks : 25 Marks

Type of Assessment Conduction : Self Learning (Home Task)

Code of Conduct:

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Question No	Conceptual Understanding 20	Model Design 25	Tool Usage 20	Analysis of Results 20	Documentation 10